

$$\begin{aligned} \hat{L}_z |l, m\rangle &= \hbar m |l, m\rangle \\ \hat{L}^2 |l, m\rangle &= \hbar^2 l(l+1) |l, m\rangle \end{aligned} \quad \left| \begin{array}{l} l = l_{\max} \\ m = -l, \dots, 0, \dots, l \end{array} \right.$$

* $\psi(r, \theta, \phi) = R(r) Y_{lm}(\theta, \phi)$ *

$$H\psi = E\psi$$

$$H = \underbrace{\left(\frac{\partial^2}{\partial r^2} \right)}_{\text{Radial}} + \underbrace{\left(\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial \phi^2} \right)}_{\text{Angular}}$$

ex: Free particle: $H = \frac{p^2}{2m} \rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$

$$H\psi = E\psi \rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

• $\psi(x) = A \cos(\alpha x) + B \sin(\alpha x)$ •

$$\rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

Aside: $\frac{d\psi}{dt} = \psi \rightarrow \int \frac{d\psi}{\psi} = \int dt$

$$\ln(\psi) = t + C \Rightarrow \psi = \underbrace{e^C}_A e^t$$

$$\psi(t) = A e^t$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi \Rightarrow \frac{\partial^2 \psi}{\partial x^2} = - \underbrace{\left(\frac{E 2m}{\hbar^2} \right)}_{\omega^2} \psi$$

* $\frac{\partial^2 \psi}{\partial x^2} = -\omega^2 \psi$ *

General solution: $\psi(x) = \underline{A e^{ix}} + \underline{B e^{-ix}}$

$$e^{ix} = \cos(x) + i \sin(x)$$

$$\psi(x) = A \cos(\omega x) + B \sin(\omega x)$$

$$\frac{\partial^2 \psi}{\partial x^2} = +\omega^2 \psi : \psi(x) = A \sinh(\omega x) + B \cosh(\omega x)$$

Back: $\psi(x) = \underbrace{A \sin(\omega x)}_{?} + \underbrace{B \cos(\omega x)}_{?}$

example:

(i) $\psi(0) = 0$

$$\psi(0) = B = 0$$

$$\psi(x) = A \sin(\omega x)$$

(ii) $\psi(L) = 0$

$$\psi(L) = A \sin(\omega L) = 0$$



$$\sin(\omega L) = 0 \Rightarrow \omega L = \sin^{-1}(0)$$

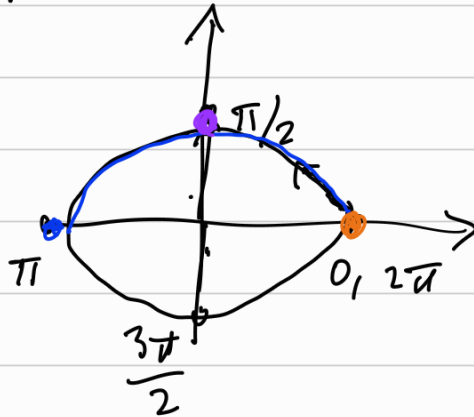
$$\underline{\omega L =}$$

* Trig fns:

cos: measure of "x"

sin: measure of "y"

Radians:



* Circle:

Perimeter: $2\pi r$

Radius: r



(i) $\boxed{r\theta = s}$

$$s = 2\pi r$$

$$r\theta = 2\pi r$$

$$\theta = 2\pi$$

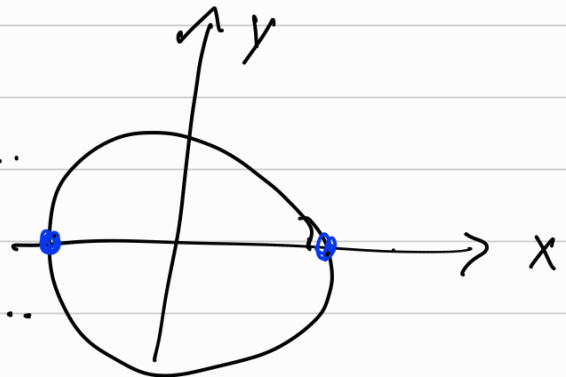
$$s = \frac{2\pi r}{2} = \pi r = r\theta \Rightarrow r = \pi$$

$$\Rightarrow \boxed{y = r \sin(\theta)} \Rightarrow \sin(\theta) = 0 = y$$

$$\sin(\theta) = 0$$

$$\theta = 0, 2\pi, \pi, -\pi, -2\pi, \dots$$

$$\boxed{\theta = n\pi}, \quad n = \dots, -2, 0, 2, \dots$$



$$\sin(\omega L) = 0 \Rightarrow \theta = \omega L = n\pi$$

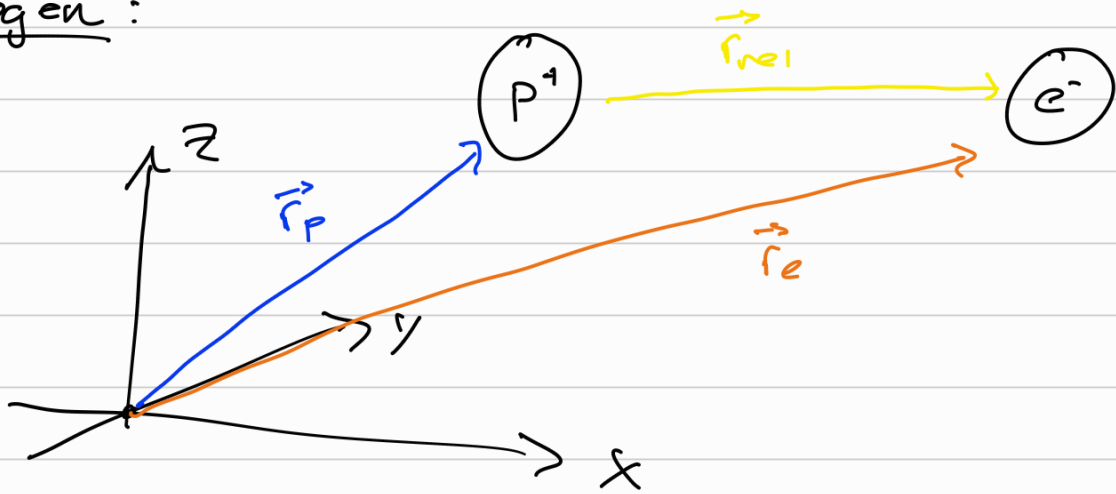
$$\underline{\omega} = n\pi / L$$

$$\omega^2 = \frac{E 2m}{\hbar^2} \Rightarrow \omega = \sqrt{\frac{E 2m}{\hbar^2}}$$

$$\sqrt{\frac{E 2m}{\hbar^2}} = \frac{n\pi}{L} \Rightarrow *$$

$$\boxed{E_n = \frac{\hbar^2 n^2 \pi^2}{L^2 2m}} *$$

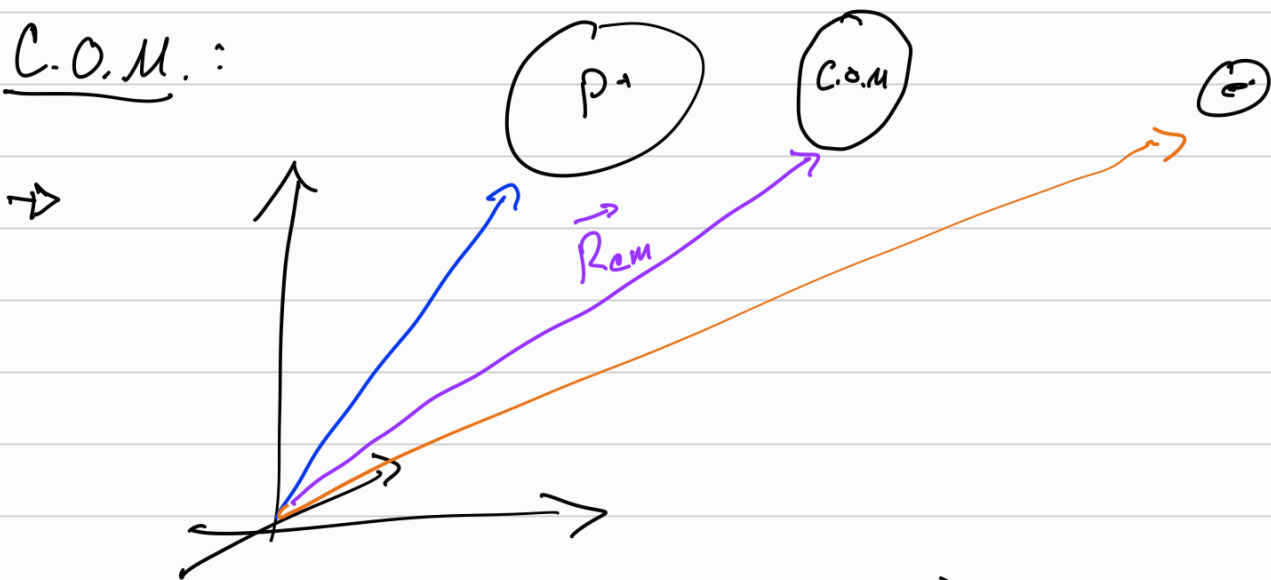
Hydrogen:



$$\vec{r}_p + \vec{r}_{rel} - \vec{r}_e = 0$$

solve for $\vec{r}_{rel} = \vec{r}_e - \vec{r}_p$

C.O.M.:



C.O.M: $\sum_i m_i \vec{r}_i = \underline{0} = \underbrace{m_{cm} \vec{r}_{cm}} + \vec{r}_p m_p + \vec{r}_e m_e$

$$\vec{r}_{cm} = \frac{1}{m_{cm}} (\vec{r}_p m_p + \vec{r}_e m_e) \quad , \quad m_{cm} = (m_p + m_e)$$

$$\vec{r}_{cm} = \frac{1}{m_p + m_e} (\vec{r}_p m_p + \vec{r}_e m_e)$$

Momentum: additive:

$$\hat{p}_{rel} = \underbrace{m_e \hat{p}_p - m_p \hat{p}_e}_{m_1 + m_2}$$

$$\vec{p}_{cm} = \vec{p}_e + \vec{p}_p$$

$$\frac{\hat{p}_{rel}}{\mu} = \frac{\hat{p}_p}{m_p} - \frac{\hat{p}_e}{m_e}$$

$$\mu = \frac{m_e m_p}{m_e + m_p}$$

$$\hat{H} = \frac{\vec{p}_p^2}{2m_p} + \frac{\vec{p}_e^2}{2m_e} + V(r_e, r_p)$$

* Solve for $\vec{p}_e$ and $\vec{p}_p$ in terms of p_{rel} and p_{cm}

$$* p_{cm} = p_e + p_p \quad \text{and} \quad \frac{p_{rel}}{\mu} = \frac{p_p}{m_p} - \frac{p_e}{m_e}$$

$$\frac{m_p p_{rel}}{\mu} = p_p - \frac{m_p}{m_e} p_e$$

$$\vec{p}_p = \frac{m_p p_{rel}}{\mu} + \frac{m_p}{m_e} p_e$$

$$p_{cm} = p_e + \left(\frac{m_p}{\mu} p_{rel} + \frac{m_p}{m_e} p_e \right)$$

Solve for $\vec{p}_e$:

$$p_{cm} = p_e \left(1 + \frac{m_p}{m_e} \right) + \frac{m_p}{\mu} p_{rel}$$

$$p_{cm} - \frac{m_p}{\mu} p_{rel} = \left(\frac{m_e + m_p}{m_e} \right) p_e$$

$$p_e = \frac{m_e}{(m_e + m_p)} \left(p_{cm} - \frac{m_p}{\mu} p_{rel} \right)$$

$$p_e = p_{cm} - \vec{p}_p$$

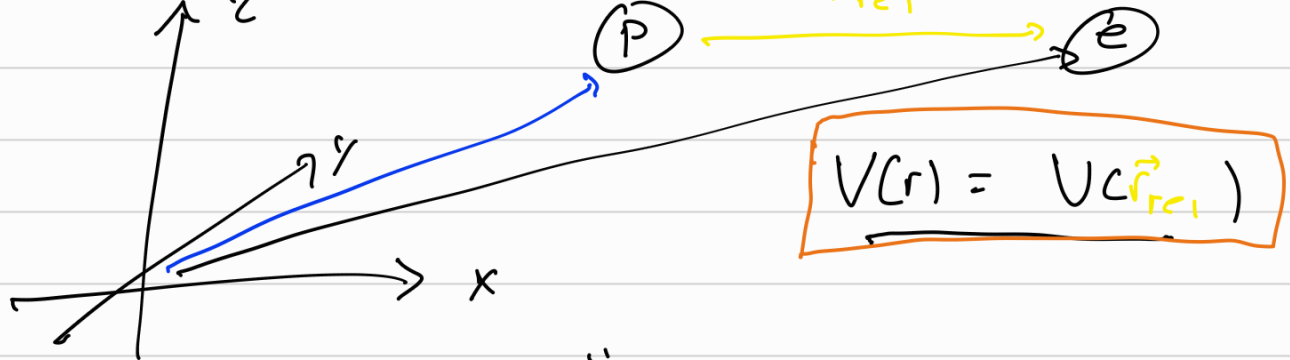
$$p_{cm} - \vec{p}_p = \frac{m_e}{(m_e + m_p)} \left(p_{cm} - \frac{m_p}{\mu} p_{rel} \right)$$

$$\vec{p}_p = p_{cm} \left(1 - \frac{m_e}{(m_e + m_p)} \right) + \frac{m_p}{\mu} p_{rel}$$

Plug into:

$$H = \frac{\vec{p}_p^2}{2m_p} + \frac{\vec{p}_e^2}{2m_e} + V(r_e, r_p) \quad (\text{homework})$$

$$\hookrightarrow H = \frac{\hat{p}_{cm}^2}{2M} + \frac{p_{rel}^2}{2\mu} + V(r_e, r_p)$$



V : Give rise to a "force"

• Free: $V = 0$:

$$\vec{F} = -\vec{\nabla} V$$

$$F = -\frac{d}{dr} V$$

Gravity: $F = -\frac{d}{dr} V$

$$F = ma = -\frac{m_1 m_2 G}{r^2}$$

$$-\frac{m_1 m_2 G}{r^2} = -\frac{d}{dr} V$$

$$\frac{d}{dr} \left(\frac{1}{r} \right) = -\frac{1}{r^2}$$

$$V = -\frac{m_1 m_2 G}{r}$$

$$F = -\frac{d}{dr} V$$

$$V_{\text{electromag}} = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$$

$$F_{\text{electromag}} = \frac{e^2}{4\pi\epsilon_0 r^2} = \frac{1}{dr} \left(\frac{e^2}{4\pi\epsilon_0 r} \right) = -\frac{d}{dr} (V)$$

$$V_{\text{electromag}} = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$$